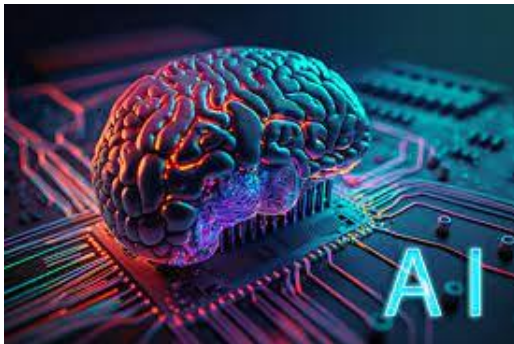


Advanced Hardware and Systems for Machine Learning

Lecture 1: Introduction



Jeff (Jun) Zhang
Assistant Professor
School of ECEE, ASU
jeffzhang@asu.edu



Administrative Details

- **A Special Topic Course: EEE/CEN 598**
- **Title: Hardware and Systems for Machine Learning**
- **Dates/Times: Tue/Thu 10:30 AM – 11:45 AM**
- **Classroom: LSEB04**

Goal of the course:

- A bridge between machine learning and hardware
- Building efficient hardware and systems for AI applications
- Learning how to do research on this topic

Acknowledgement: Mohamed Abdelfattah, Joel Emer, Song Han, Lizy John, Jae-sun Seo, Sophia Shao, Vivienne Sze

Instructor

- **Instructor: Jeff (Jun) Zhang**
- **Office: ~~ISTB4-467~~---> GWC 242**
- **Email: jeffzhang@asu.edu**
- **Office hours:**
 - Tue/Thu 9:45 AM – 10:25 AM



Background of Your Instructor

- **Education:**
 - Ph.D., Electrical Engineering, New York University, 2020.
- **Work Experience:**
 - Postdoc, Harvard University, 2020–2023.
 - Research Intern, Microsoft Research and Samsung, 2018
- **Faculty at ASU:**
 - Currently an Assistant Professor at ECEE, since January 2023.
- **Research Interests:**
 - Computer Architecture, VLSI Chip Design, EDA, Embedded Systems
 - Energy Efficient and Robust Deep Learning
 - Website: <https://search.asu.edu/profile/4346755>
 - We always love to work with motivated students!

Course Information

ASU Canvas

- Announcements
- Lecture slides
- Homework/lab assignments
- Readings, paper discussion and reviews
- ED discussion
- Grades

Discussions and Emails

- **Discussions at “Ed Discussion” (accessible via Canvas)**
 - All questions that are not personal in nature should be posted at the Ed Discussion forum. These include all technical questions regarding lecture notes, assignments, CAD tools, server access, etc.
 - You may get faster response from the instructor, graders, or your classmates.
 - Similar questions and answers can be shared for everybody.
 - Please ask yourself why you cannot post your questions on Canvas before sending emails to the instructor or the grader.
 - Up to **5% bonus points** will be provided to the students who contributed the most in the Ed Discussion forums throughout the semester.
- **Emails**
 - The subject of all emails to the instructor should start with “EEE 598 ML Hardware: XXX”.

Course Topics (Subject to Change)

■ **Core topics:**

- Deep Neural Network Basics
- Model optimization techniques: quantization, pruning, etc.
- Architectural optimizations: performance, energy efficiency, etc.
- Custom hardware: systolic array, CGRA, other accelerators.
- Compilation techniques: tiling, data flow, mapping, scheduling
- AutoML: Neural architecture/hardware co-design
- TinyML: resource constrained DNN optimization
- Systems for ML: ML inference serving system
- And more...

Prerequisites

- EEE 425/525, Computer Architecture
- Familiarity with digital IC design flow (e.g., logic synthesis)
- Introductory ML or deep learning
- Familiarity with C/C++, Python/PyTorch, Verilog
- Prior project experience will be a plus
 - FPGA, ASIC development
 - ML application/algorithm development

Assessment (Subject to Change)

- **Paper reading: 15%**
 - Reading assignments and paper review (posted on Canvas)
- **Homework: 25%**
 - Three practical hands-on assignments
 - Done solo
- **Research project: 50%**
 - Do great research (hard working)!
- **Class participation and discussion: 10%**
 - Get involved in research

Paper Reading and Paper Review

- **Reading**: an essential component of research
- One or two papers per week, usually due before a lecture
- Think about the following when you are reading:
 - What is the **biggest challenge** and **motivation** of this work?
 - What is the **proposed methodology** and why they make (or not make) sense
 - How did the work **evaluate** the proposed solutions?
 - What tools, datasets, experimental results to run
 - How do you like the problem, idea, evaluation of the paper? (**strength and weakness**)
 - What can be done/improved as the **future work**
 - What **questions** you have from the reading

More details on the Project

- **Open ended research project: propose your own**
 - E.g., new hardware design: ML Algorithm, RTL → layout (GDS)
 - E.g., hardware-aware ML: on device AI, GPU accelerated AI
 - E.g., new ML system
- **Team project**
 - Default group size: 3 students
 - Team up with your classmates soon!
 - *Grade each other's contribution.*
- **Requirement:**
 - Intermediate report (IEEE conf format, due: week 13)
 - Final presentation (week 15 & 16)
 - Final report (due: week 16)
 - At least 6 pages, double-column
 - Potential submission to IEEE/ACM conferences

Suggested CMOS Technology and CAD Tools

- **7nm educational PDK**
 - Developed by ASU (led by Prof. L. Clark) & ARM
- **RTL (Verilog) Simulation: ModelSim, NCVerilog, etc.**
- **Synthesis: Synopsys Design Compiler**
- **APR (Automatic Place & Route): Cadence Innovus**
- **Power simulation: PrimeTime PX**

Computing Resources

- **EDA tools on Apporto (enabled in course Canvas)**
- **ASU Research Computing Sol Supercomputer**
 - GPUs: A30, A100; CPUs; TB memories.
 - Jupyter notebooks for deep learning.

- **Cloud CPUs/GPUs:**

- Google Colab

Plan Tier	Typical GPUs Available	Best Used For	Key Features & Limits
Free Tier	NVIDIA T4 (~12GB VRAM)	Training small/mid-sized Transformers, fine-tuning BERT/Whisper, CNN image classification.	Subject to dynamic availability; 12-hour max session lifetime; idle timeouts.
Colab Pro	NVIDIA V100, A100	Large language model (LLM) fine-tuning, heavy deep learning pipelines.	100 compute units/month, faster GPU allocation, expanded memory.

- Amazon SageMaker Studio Lab (free 12-hour CPU session or 4-hour GPU session)

Expected Grade Distribution

- Letter grade depends on the **relative** distribution, with + and – (μ : average; σ : standard deviation)
 - A+ : 3~4 students
 - A : $> \mu + 0.5\sigma$
 - A- : $> \mu$
 - B+ : $> \mu - 0.5\sigma$
 - B : $> \mu - 1.0\sigma$
 - B- : $> \mu - 1.5\sigma$
 - C : $> \mu - 2.0\sigma$
 - D : else

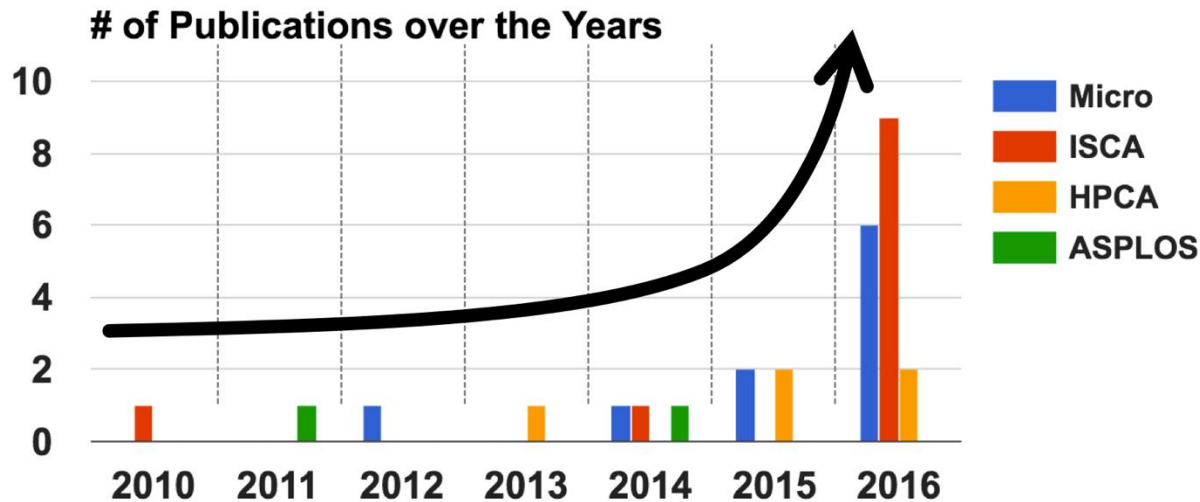
What is Good Research

- Be productive: manage your time
- Be active: take the lead, self-driven
- Reading the state-of-the-art research papers
- Critical thinking and discussion
- Get your hands dirty: implementing your ideas
- **The Zen of Research - Tips for New Researchers in EECS**
 - Prof. Todd Austin, University of Michigan
 - <https://www.youtube.com/watch?v=4eUT0kAg9SA>

Top Conferences in Our Community

- **Arch: ISCA, MICRO, ASPLOS, HPCA, MLSys**
- **EDA: DAC, ICCAD, DATE, ASP-DAC, ...**
- **Circuits: ISSCC, VLSI symposium, ESSCIRC, CICC**
- **VLSI Test: ITC, VTS, ...**
- **AI/ML: NeurIPS, ICCV, CVPR, ICML, AAAI, ...**

Deep Learning Publications at Architecture Conferences



HPCA 2023



Session 1A: Neural Networks and Accelerators 1

Session 4A: Neural Networks and Accelerators 2

Session 7A: Neural Network and Accelerators 3

Also new conferences focusing on addressing compute challenges for ML/DNN, e.g., MLSys, AICAS, TinyML

Size and Emer

Notes/Policy

- **HomeWorks / projects should be electronically submitted at Canvas.**
- **Late submissions will have hefty penalty (point deduction)**
 - Make first submission much ahead of deadline
 - Incrementally update your submission

Academic Integrity

- In the “Student Academic Integrity Policy” manual, ASU defines “Plagiarism” as using another's words, ideas, materials or work without properly acknowledging and documenting the source. Students are responsible for knowing the rules governing the use of another's work or materials and for acknowledging and documenting the source appropriately. Academic dishonesty, including inappropriate collaboration, will not be tolerated and will be reported to the Dean of the school. There are severe sanctions for cheating, plagiarizing and any other form of dishonesty.
- For example, final project score can be zero.

Artificial Intelligence (AI)

Artificial Intelligence

“The science and engineering of
creating intelligent machines”

John McCarthy, 1956

Eyeriss tutorial

Machine Learning (ML)

Artificial Intelligence

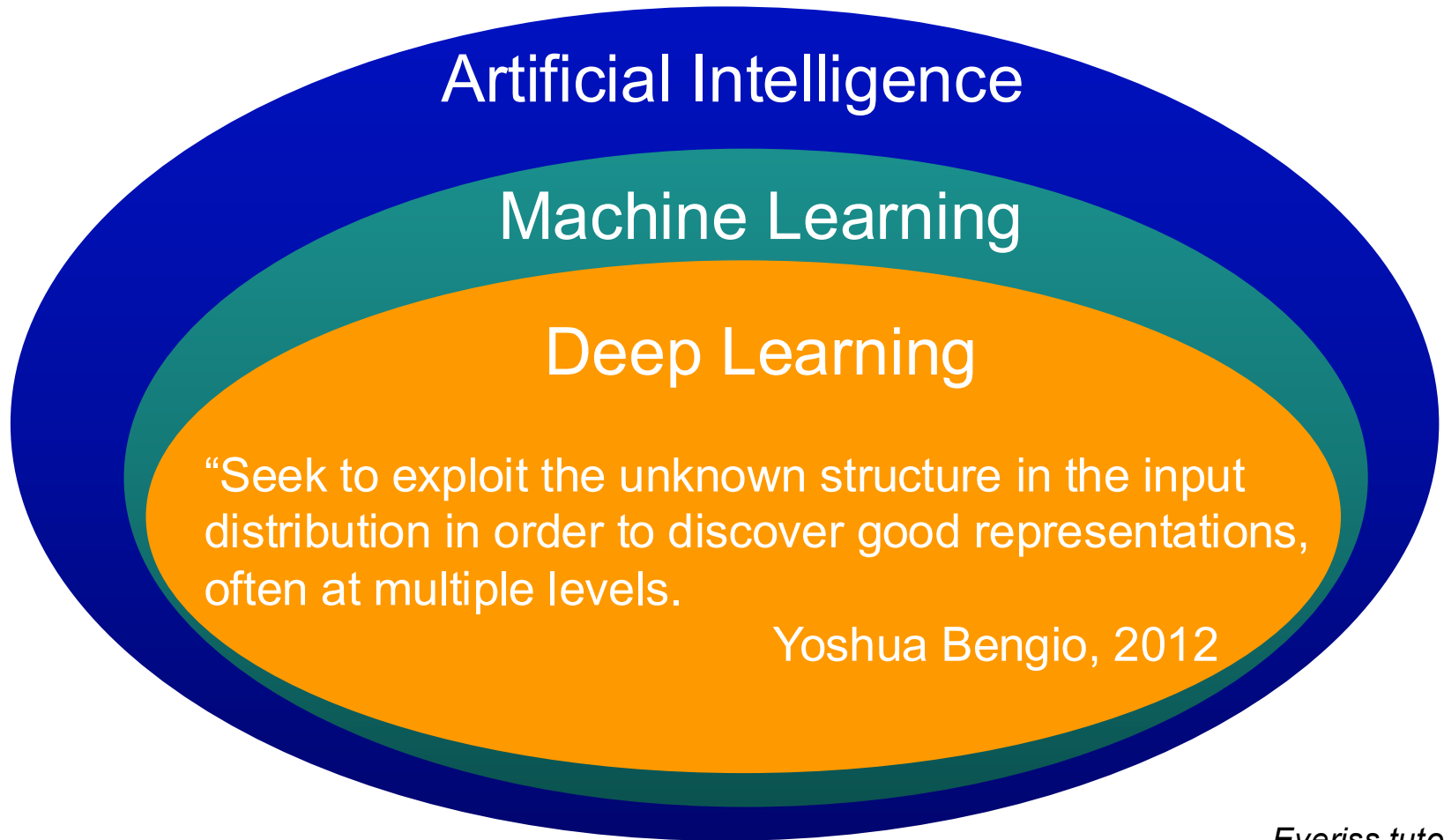
Machine Learning

“Field of study that gives computers the ability to learn without being explicitly programmed.”

Arthur Samuel, 1959

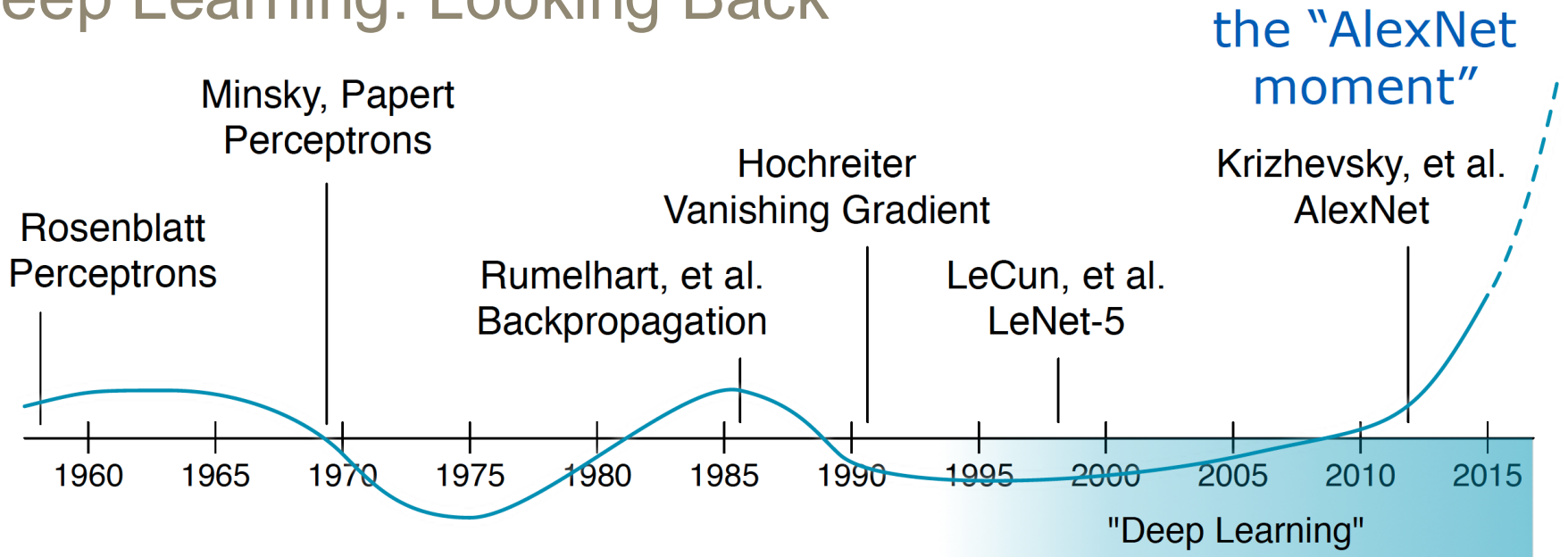
Eyeriss tutorial

Deep Learning (DL)



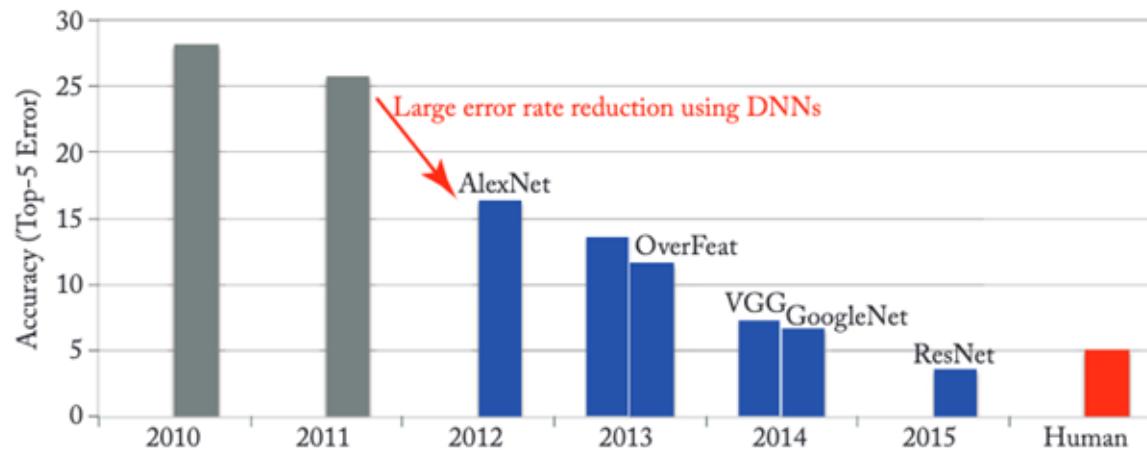
Eyeriss tutorial

Deep Learning: Looking Back

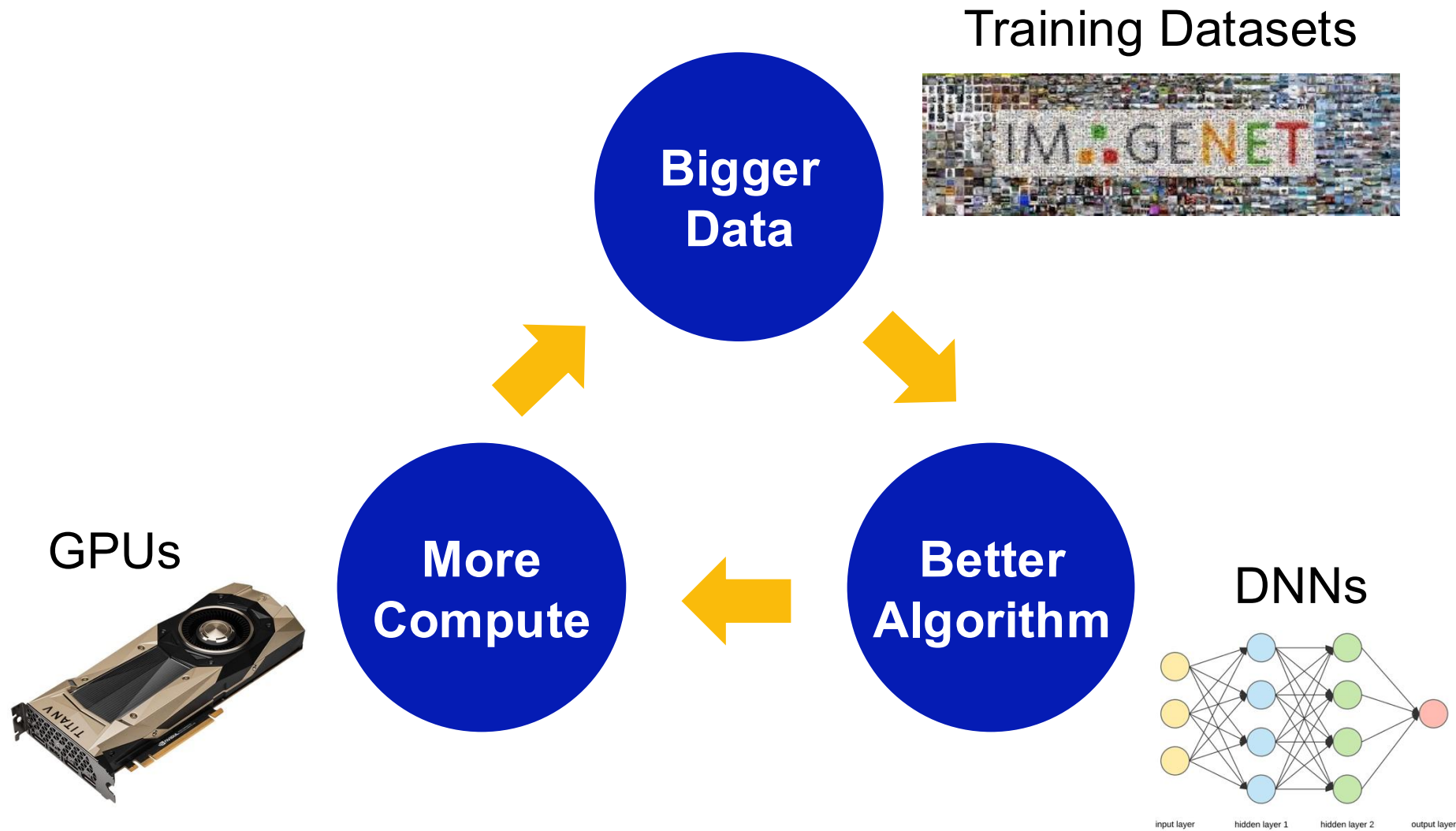


— Hype Curve

[Reagen *et al.*, Deep Learning for Computer Architects, 2017]



The Success of Deep Learning



Where is DL/ML being Used Today?

Search **Translation** **Ads**

Face tagging

News Feed

Long tail of applications

- Content understanding
- Speech recognition
- Objectionable content detection
- Fraudulent account detection
- Content integrity
- Sentiment analysis

AI-enabled Applications

Recommender Systems



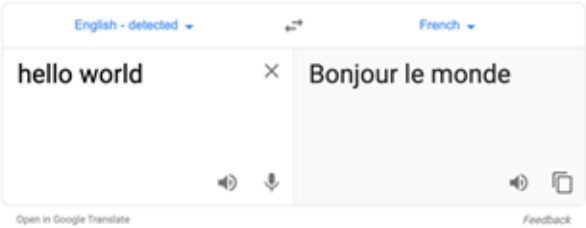
Vision



Speech



Natural Language Processing



Natural Language Processing

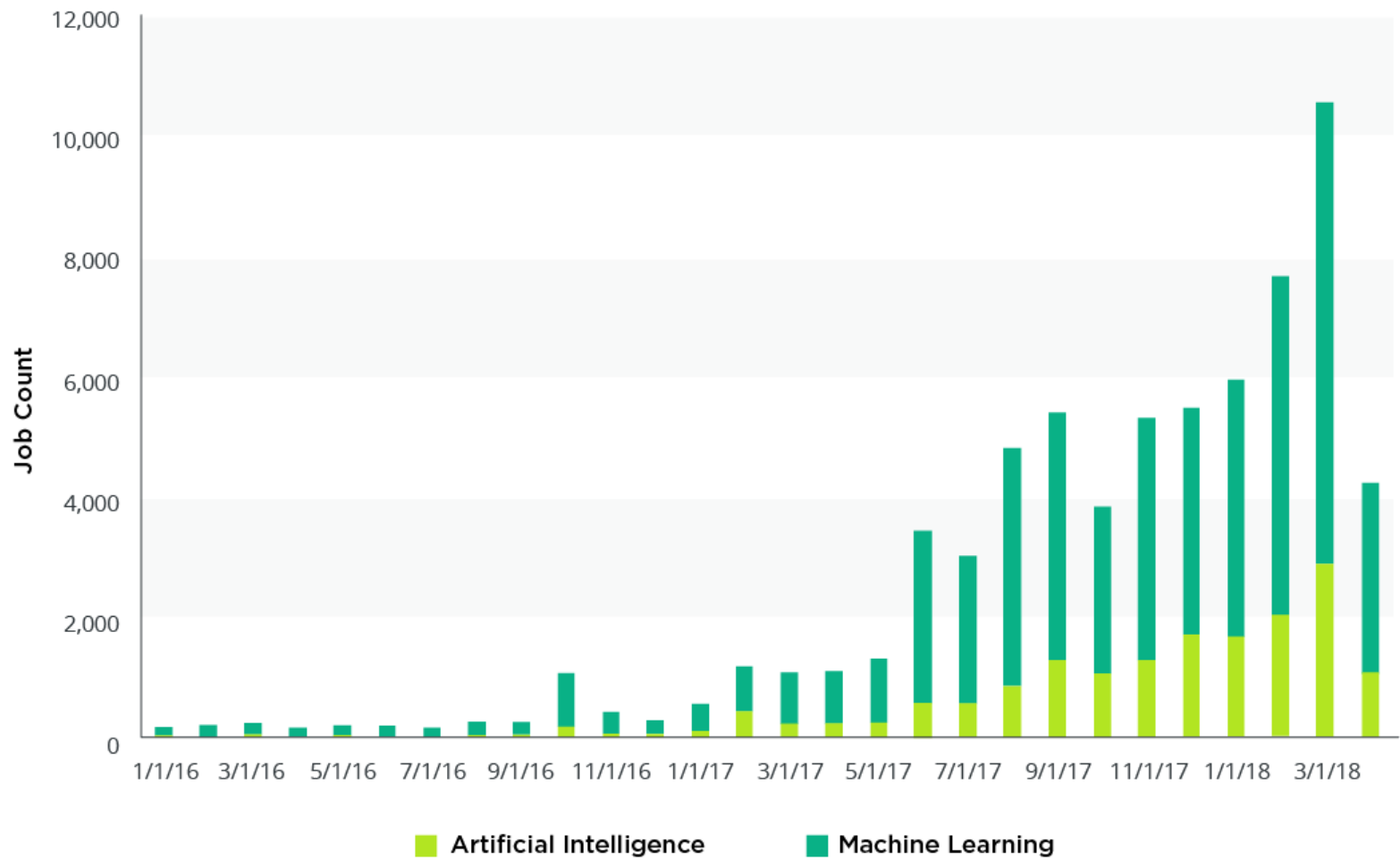


Robotics and Drones



AI/ML Jobs

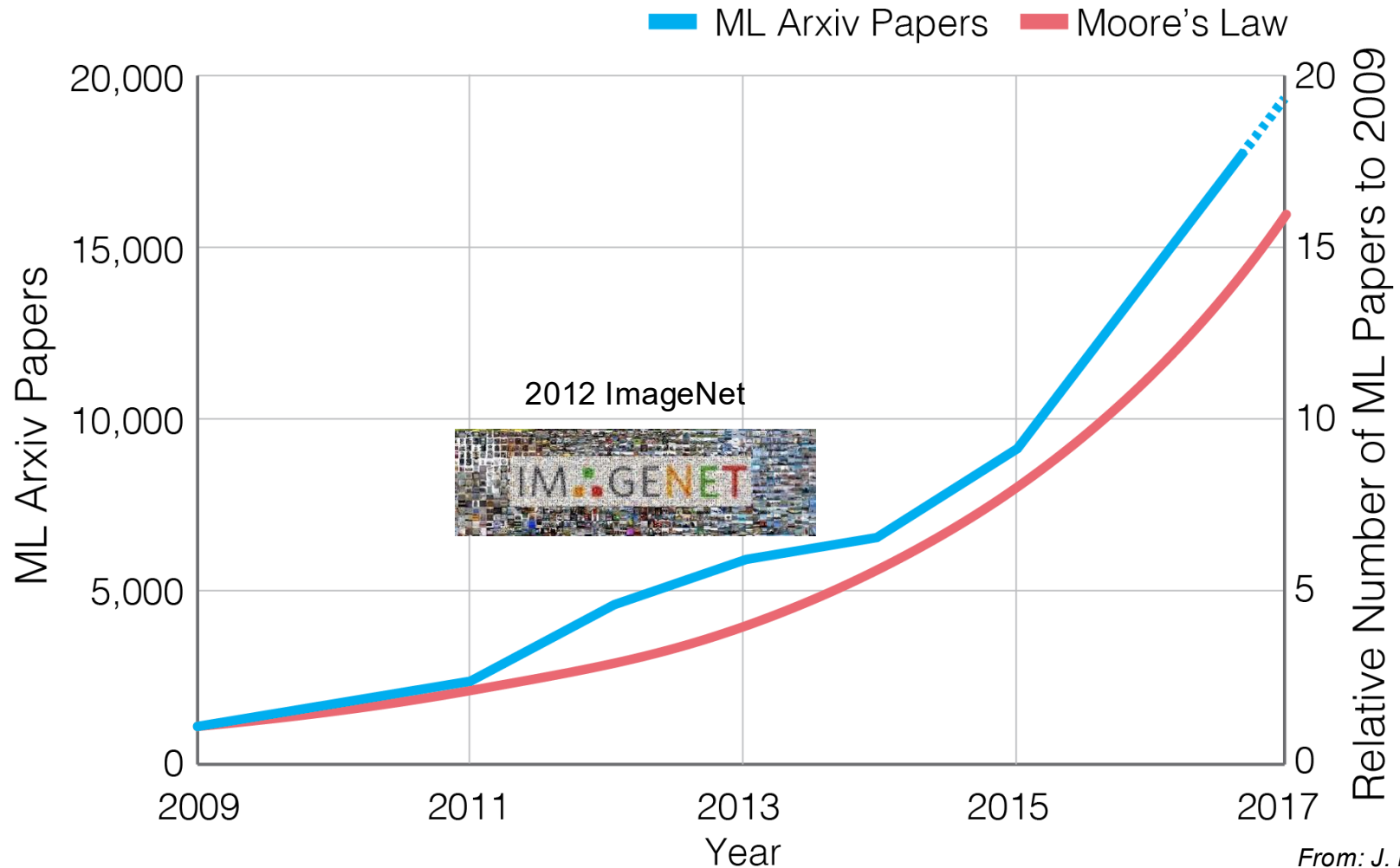
Total Job Postings Referencing AI Skills



From: ZipRecruiter



Deep Learning Research Papers



From: J. Dean, 2018

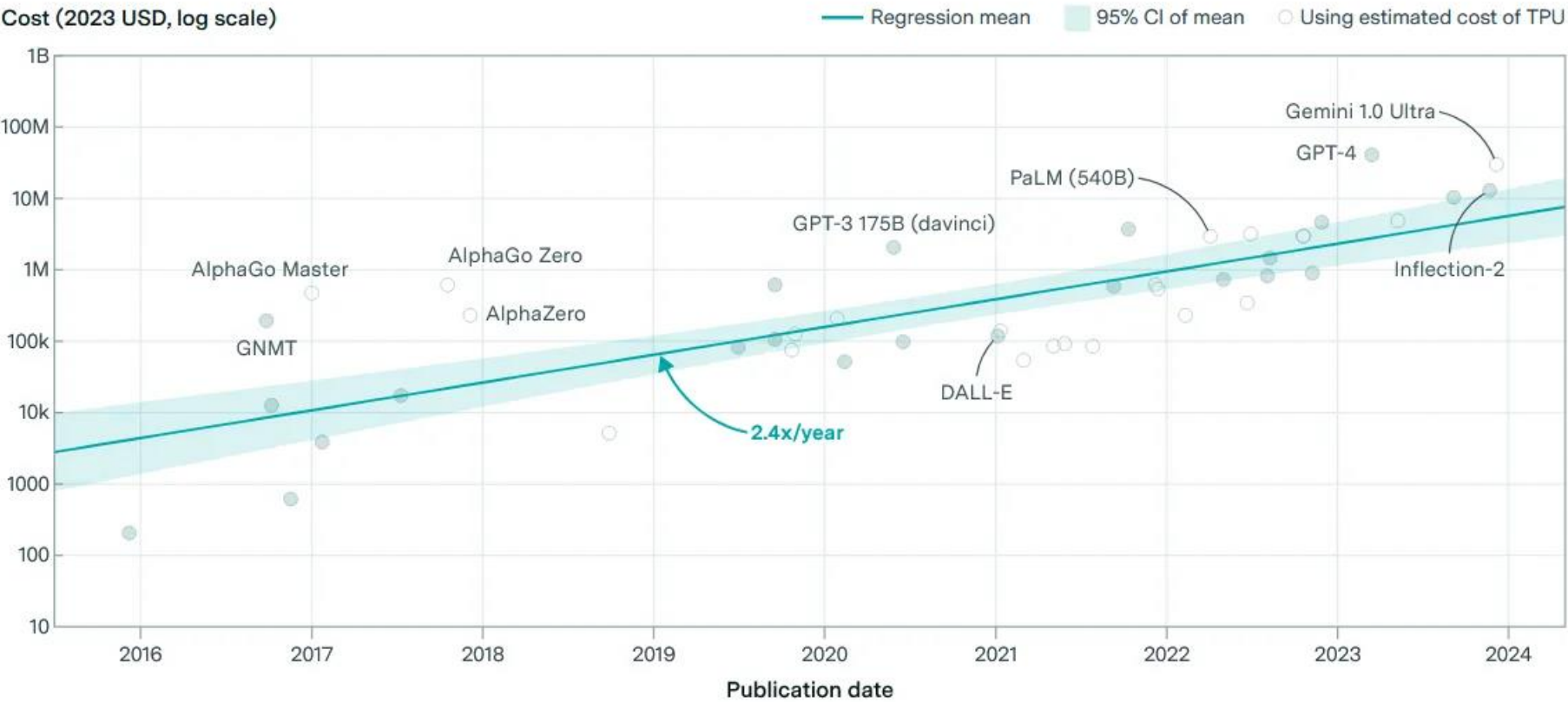
Scaling Deep Learning Models



From: Computing Power and the Governance of Artificial Intelligence

Scaling Deep Learning Models: Cost

Amortized hardware and energy cost to train frontier AI models over time



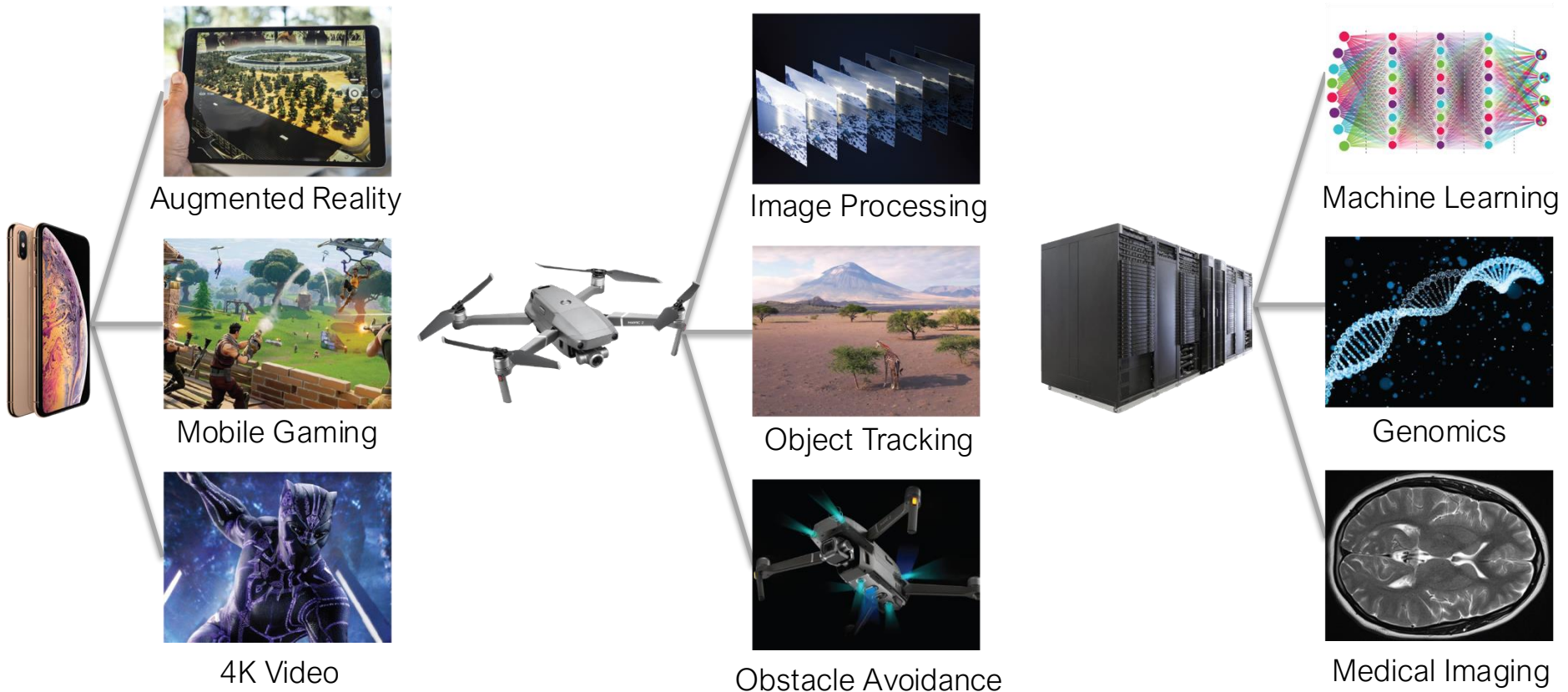
Scaling Deep Learning Hardware

Table 1: 90-epoch training time and single-crop validation accuracy of ResNet-50 for ImageNet reported by different teams.

Team	Hardware	Software	Minibatch size	Time	Accuracy
He <i>et al.</i> [5]	Tesla P100 × 8	Caffe	256	29 hr	75.3 %
Goyal <i>et al.</i> [4]	Tesla P100 × 256	Caffe2	8,192	1 hr	76.3 %
Codreanu <i>et al.</i> [3]	KNL 7250 × 720	Intel Caffe	11,520	62 min	75.0 %
You <i>et al.</i> [10]	Xeon 8160 × 1600	Intel Caffe	16,000	31 min	75.3 %
This work	Tesla P100 × 1024	Chainer	32,768	15 min	74.9 %

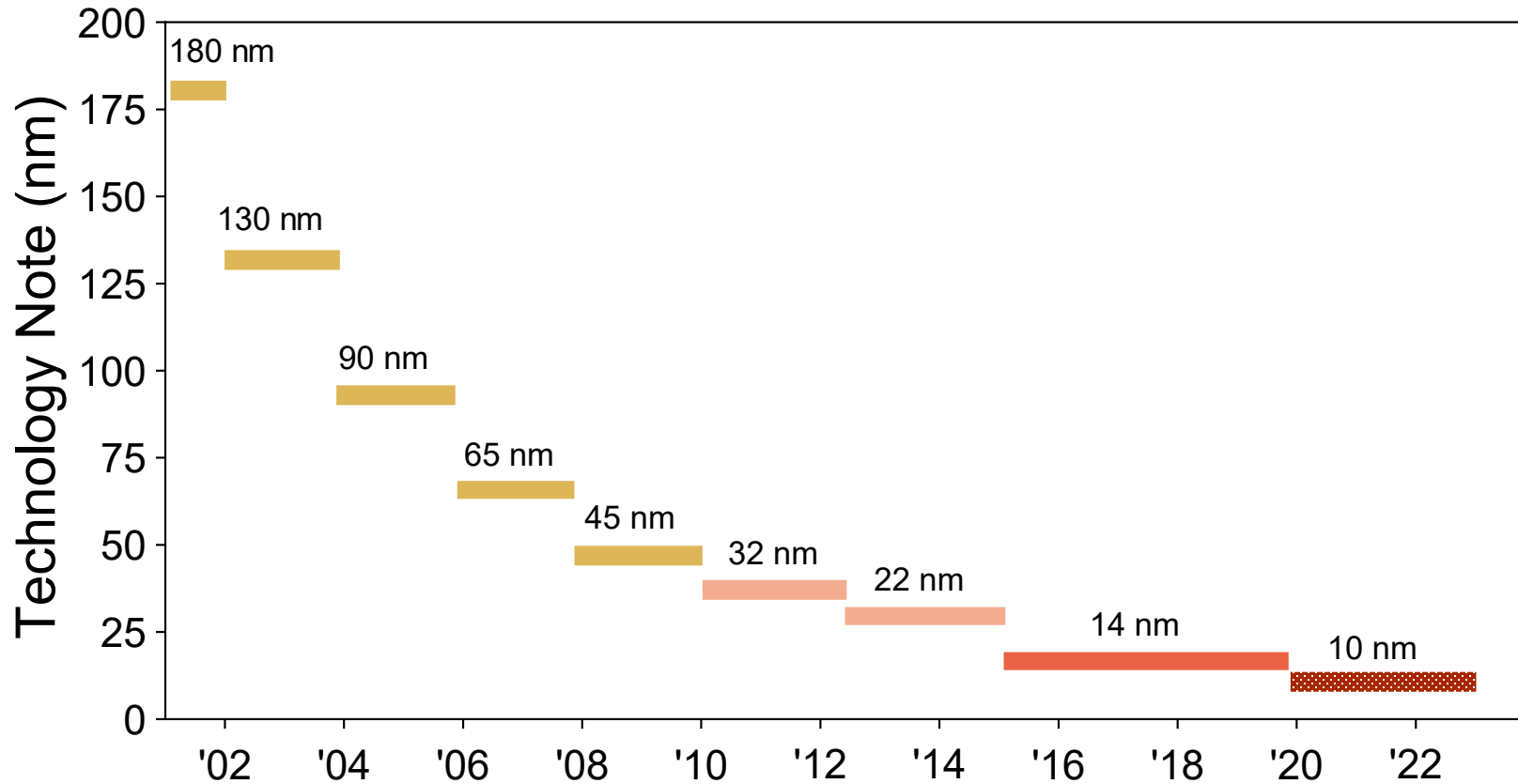
From: Extremely Large Minibatch SGD: Training ResNet-50 on ImageNet in 15 Minutes, 2017

Diverse Hardware for Machine Learning



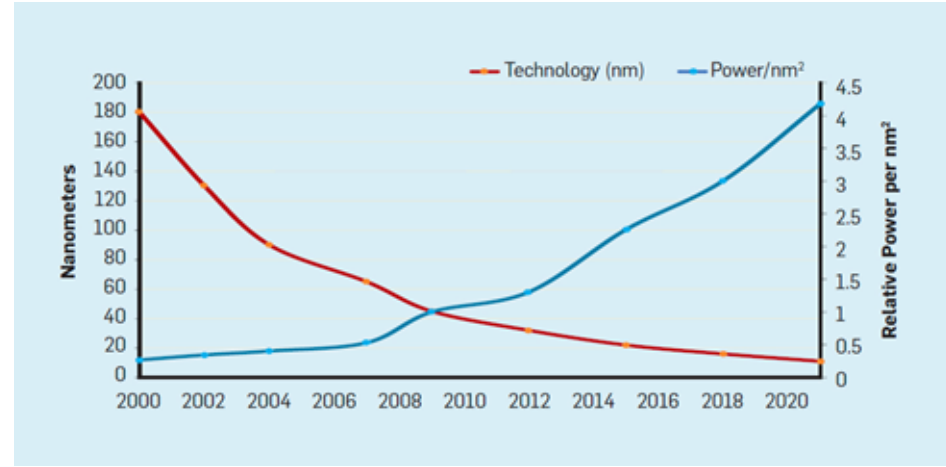
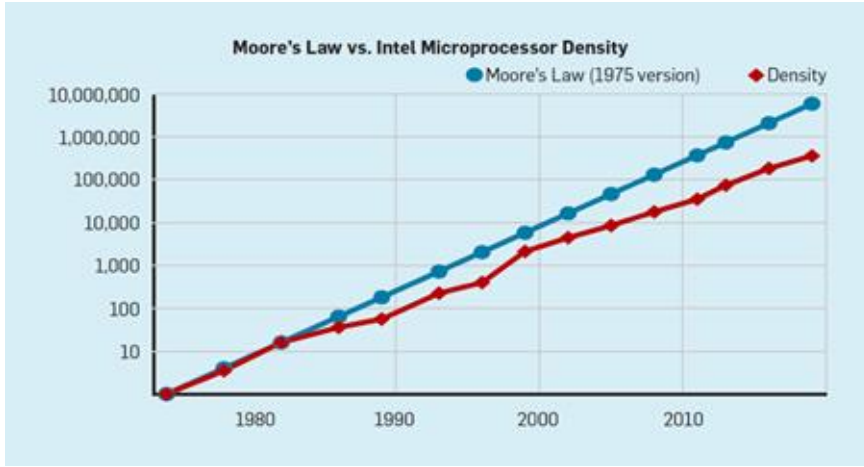
Increasing Demand for Computing: from **edge** to the **cloud**

Moore's Law is Slowing Down



* <http://www.anandtech.com/show/9447/intel-10nm-and-kaby-lake>

What could we do?



<https://cacm.acm.org/magazines/2019/2/234352-a-new-golden-age-for-computer-architecture>



- For the past 4 decades, we can put more transistors on a chip and get exponentially more performance → this is ending.
 - And it is ending at a time when the demand for computational power is skyrocketing!
- Can't rely on technology for performance
 - Need new hardware architectures
 - need new software stacks to program these architectures

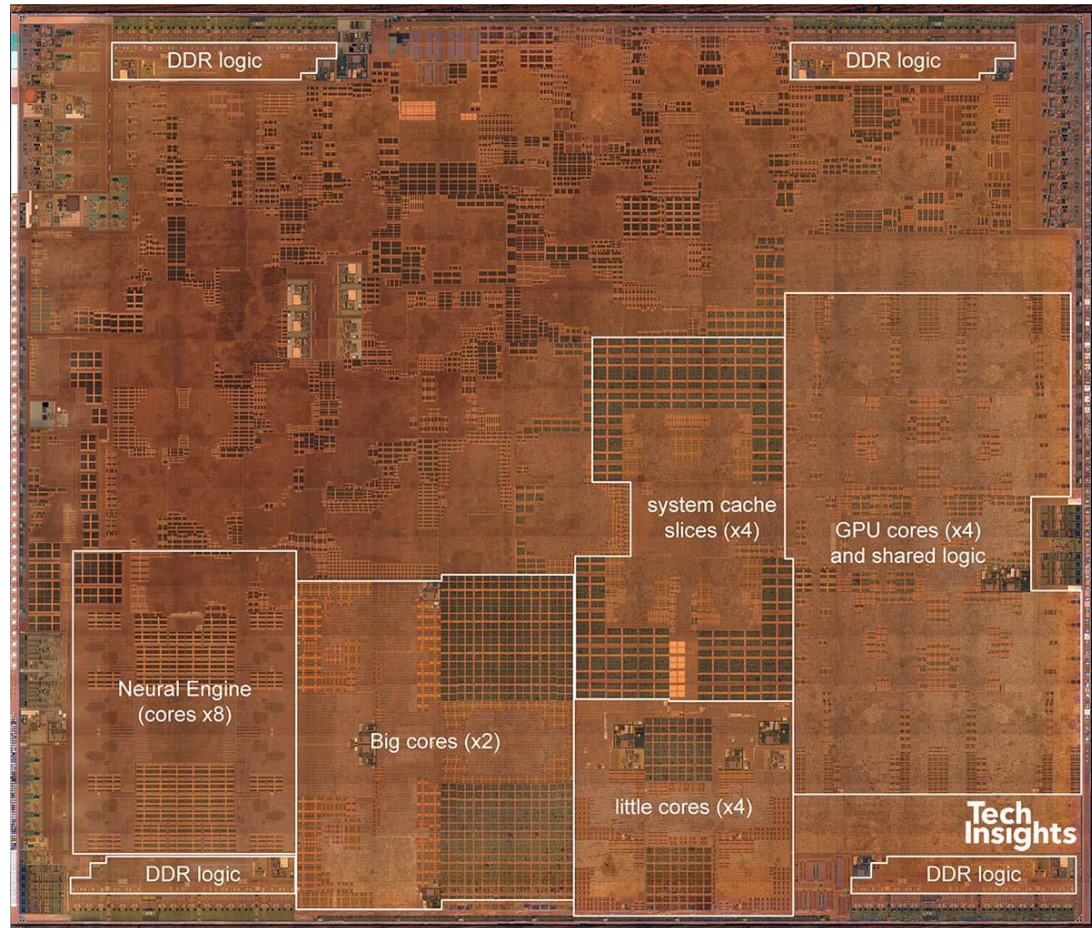
Domain Specific Hardware!

Domain-Specific Accelerators

Customized hardware designed for a domain of applications.



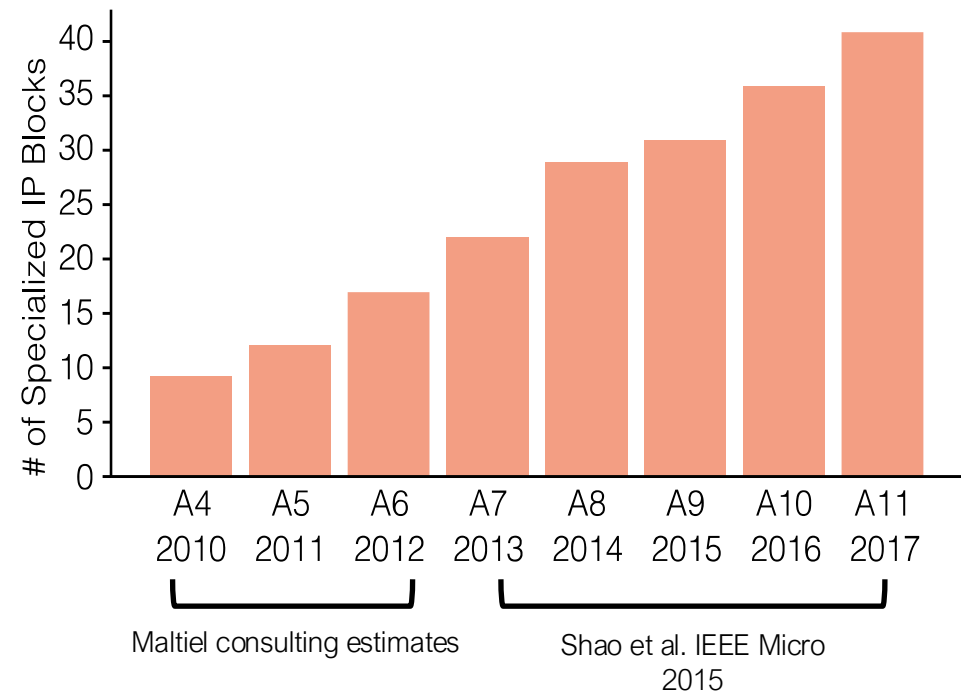
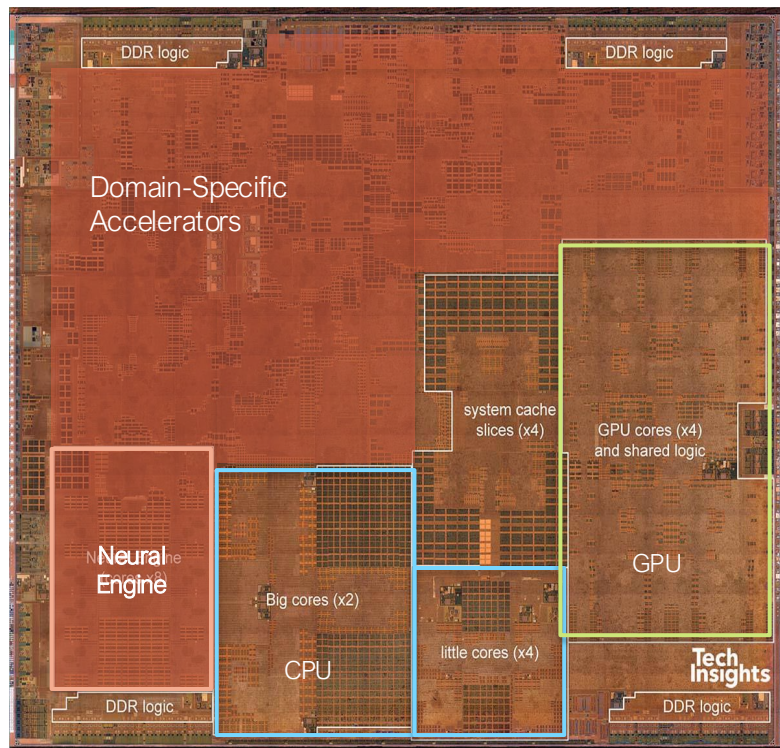
iPhone
XS, 2018



* TechInsights.com Apple iPhone XS teardown

Apple SoCs

Customized hardware designed for a domain of applications.



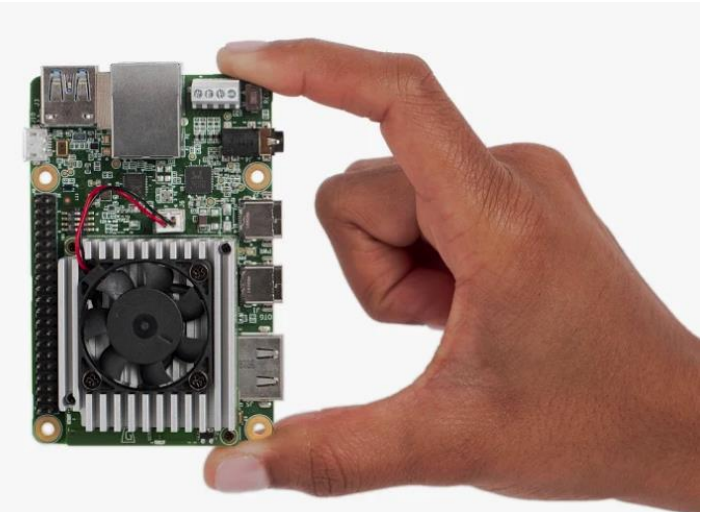
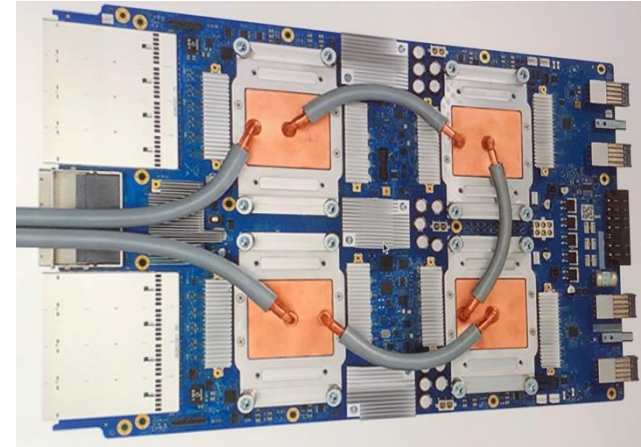
Google TPU

- **Systolic-array-based architecture**

- V1: Inference only
- V2: Training with bfloat
- V3: 2x powerful than v2

- **Edge TPU**

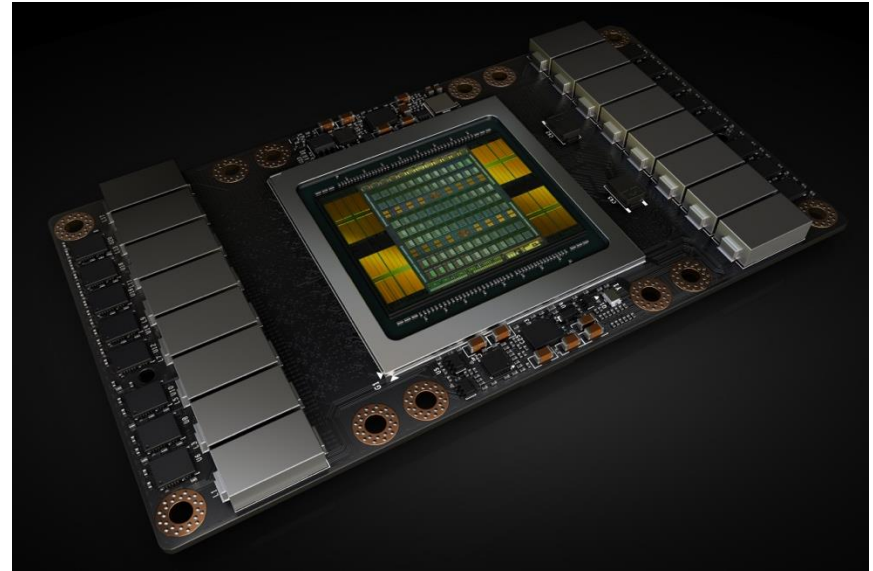
- Coral Dev Board
- 4 TOPS
- 2 TOPS/Watt
- Supports TensorFlow Lite



NVIDIA GPU

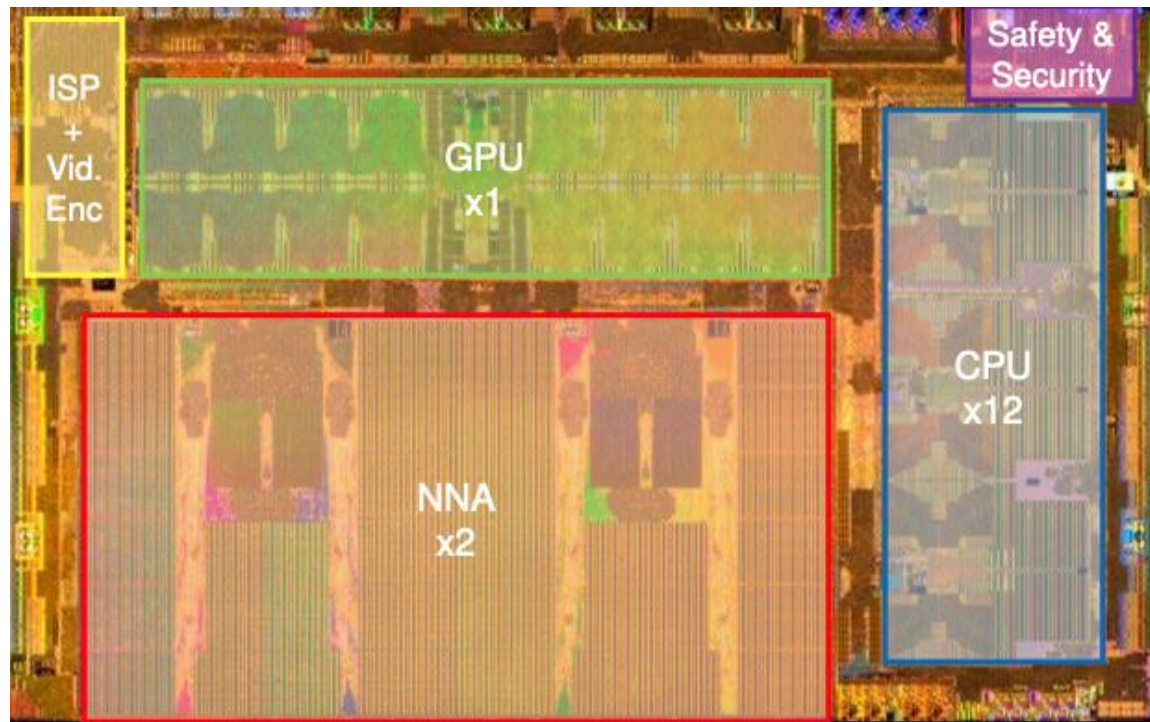
NVIDIA Volta V100 GPUs

- 21 billion transistors
- Die size 815 mm²
- TSMC 12 nm FinFET
- 15.7 TFLOP/s of single precision (FP32) performance
- 125 Tensor TFLOP/s of mixed-precision matrix-multiply-and-accumulate
TDP 300W



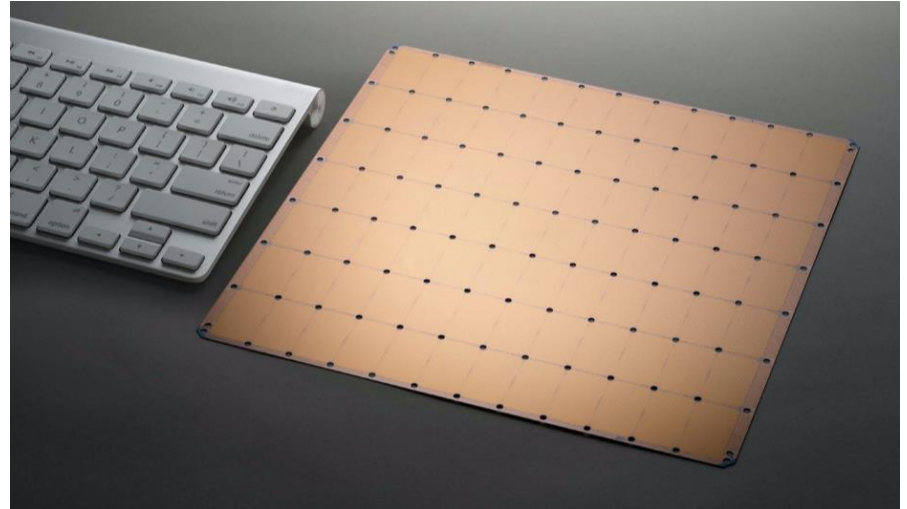
Tesla Full Self Driving Chip

- 2 independent instances
- 2Ghz+ Design
- 32 MB SRAM / instance
- 96*96 MACs

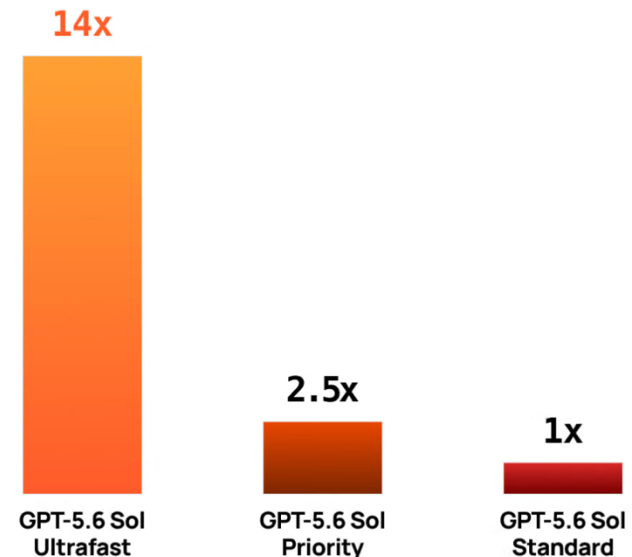


Cerebras Wafer Scale Computing Engine

- Largest Chip Ever Built!
- 46,225 mm² silicon
- 1.2 trillion transistors
- 400,000 optimized AI cores
- 18 GB of on-chip memory
- TSMC 16nm process



GPT-5.6 Sol Ultrafast is up to **14x faster** than GPT-5.6 Sol Standard

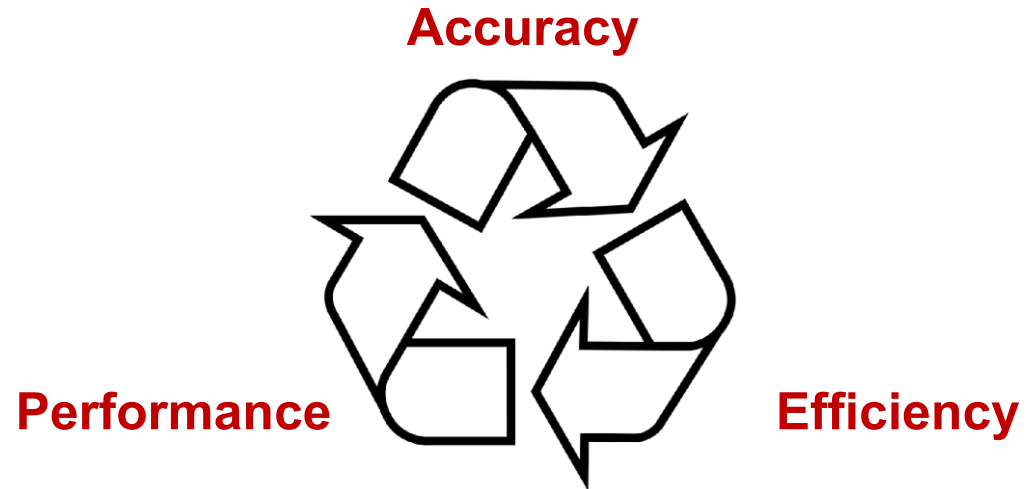
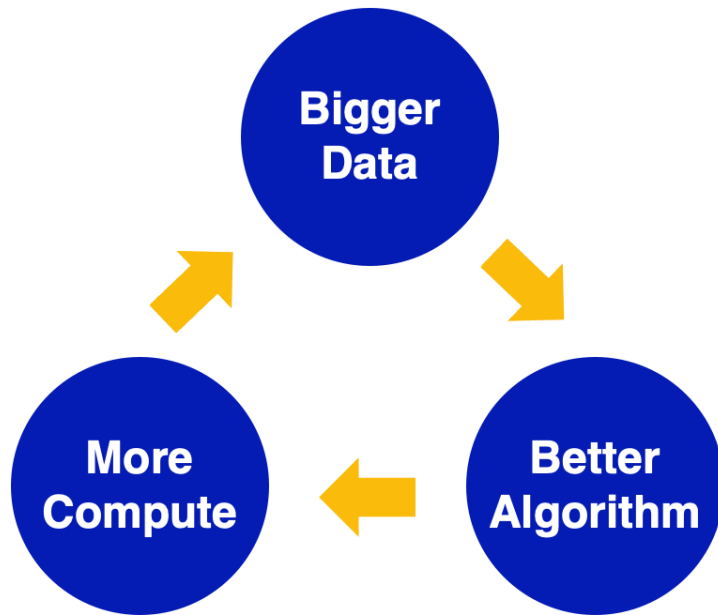


Domain-Specific Accelerators --- What do you observe?

1. [Esperanto](#) Accelerating ML Recommendation with over a Thousand RISC-V/Tensor Processors on Esperanto's ET-SoC-1 Chip, 2021
2. [Cerebras](#) Wafer Scale Engine 2 WSE-2 at Hot Chips 33, 2021
3. [IBM](#) Unveils On-Chip Accelerated Artificial Intelligence Processor, 2021
4. [Qualcomm](#) Cloud AI 100 AI Inference Card at Hot Chips 33, 2021
5. [SambaNova](#) SN10 RDU: Accelerating Software 2.0 with Dataflow, 2021
6. [Tesla's](#) AI chip REVEALED! (Project Dojo), 2021
7. The [Apple](#) Neural Engine, 2021
8. An overview of EC2 Inf1 Instances featuring [Amazon](#) Inferentia chips, 2020
9. [Meta](#) open-sources hardware for AI model training and inference, 2019, and [AR/VR](#) chip, 2022
10. [Habana](#), the AI chip innovator, promises top performance and efficiency, 2019
11. [Google](#) In-Datacenter Performance Analysis of a Tensor Processing Unit, 2017

This course --- Building you own system!

- Better algorithms/applications
- Better runtimes/software
- Better hardware



Further Reading

- 🔗 **A new Golden age for Computer Architecture**
- 🔗 2018 Turing Lecture

turing lecture

DOI:10.1145/3282307

Innovations like domain-specific hardware, enhanced security, open instruction sets, and agile chip development will lead the way.

BY JOHN L. HENNESSY AND DAVID A. PATTERSON

A New Golden Age for Computer Architecture